

Anant A. Joshi

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Summary

- As a control theorist and applied mathematician, my foundation is in **stochastic differential equations, stochastic optimal control (SOC), and filtering**. I am interested in problems in **sampling, SOC, generative models, and optimization**. My research focusses on **constructing sampling-based algorithms for SOC and reinforcement learning (RL) using interacting particle systems**.

Education

Ph.D. in Mech. Engg., Univ. of Illinois Urbana-Champaign (GPA 4.0/4.0, Advisor: Prof. Prashant Mehta)

Jan. 2021 - present

- Awarded: Grainger Engineering Graduate Fellowship

B.Tech. in Mech. Engg. and M.Tech. in Systems and Control Engg., IIT Bombay, India (GPA 9.44/10)

Aug. 2015 - July 2020

- Awarded: Best student paper award, 29th AAS/AIAA Space-Flight Mechanics Meeting, 2019

Patents

- M. Benosman, S. Mowlavi, **A. A. Joshi**, "System and Method Suitable for Synthesizing a Control Policy for a Mechanical System", USPTO App. 19/273,383

Experience

dual-EnKF Algorithm For Fast Simulation-Based Reinforcement Learning

UIUC

RESEARCH WITH PROF. P. G. MEHTA, PROF. A. TAGHVAEI, PROF. S. P. MEYN

Jan 2021 - Dec 2024

- Developed a simulation based algorithm (dual-EnKF) for stochastic optimal control and RL in continuous-time continuous-state systems.
- Represented value function as a probability density and deployed the Ensemble Kalman Filter (EnKF) to sample efficiently from it.
- Numerically benchmarked dual-EnKF against policy gradient method, demonstrating upto 100x computational speedup.

Industry Internship: dual-EnKF for Data Driven Robust Control of PDEs

Mitsubishi Electric Research Labs

RESEARCH INTERN WITH DR. MOUHACINE BENOSMAN AND DR. SAVIZ MOWLAVI

June 2024 - August 2024

- Developed a data-driven, robust control algorithm for nonlinear systems governed by partial differential equations. Combined an optimal control estimated using dual-EnKF with a robust control obtained from Lyapunov redesign.
- Validated the controller's performance and robustness through extensive simulations on the heat equation and Burgers' equation.

Error Analysis of Sampling Algorithms for Stochastic Control

UIUC

RESEARCH WITH PROF. P. G. MEHTA AND PROF. A. TAGHVAEI

Jan 2025 - ongoing

- Developed an interacting particle-based sampling algorithm to numerically solve discrete-time stochastic optimal control. Analyzed the algorithm's approximation error and performance, and benchmarked it against traditional importance sampling methods.
- Highlighted scalability by demonstrating that the proposed algorithm avoids "curse of dimensionality" seen in importance sampling.

Observer Design for Systems on Smooth Manifolds with Lie Group Symmetry

IIT Bombay, India

RESEARCH PROJECT WITH PROF. R. N. BANAVAR AND PROF. D. H. S. MAITHRIPALA

Aug. 2018 - Jul. 2019

- Leveraged inherent geometric symmetry in nonlinear mechanical systems to simplify design observer into two smaller subsystems.

Uncertainty Quantification in Dynamical Systems with Invariants (AAS/AIAA Best paper)

University of Texas at Arlington

RESEARCH INTERNSHIP WITH PROF. K. S. SUBBARAO

May 2018 - July 2018

- Modeled impact of Gaussian white noise on dynamical systems to quantify uncertainty in conserved quantities (invariants).
- Characterized the temporal evolution of the statistical moments of invariants. Theoretical results were verified with Monte Carlo simulation.

Leadership Roles

Head, Attitude Determination and Control Subsystem, Student Satellite Project

IIT Bombay

- Responsible for envisioning and executing technical work in the 10 member subsystem. Regularly presented progress review to the faculty mentors.
- Laid out Quality-Assurance practices to ensure categorical documentation of project progress.

Selected Publications

- A. A. Joshi**, S. Mowlavi, M. Benosman, "A Dual Ensemble Kalman Filter Approach to Robust Control of Nonlinear Systems: An Application to Partial Differential Equations", accepted to IEEE Conference on Decision and Control (CDC), 2025
- A. A. Joshi**, H.-S. Chang, A. Taghvaei, P. G. Mehta, S. P. Meyn, "Interacting Particle Systems for Fast Linear Quadratic RL", Learning for Dynamics & Control Conference (L4DC), 2024
- P. Katdare, **A. A. Joshi**, K. Driggs-Campbell, "Towards Provable Log Density Policy Gradient", Transactions on Machine Learning Research (TMLR), 2024
- A. A. Joshi**, D. Chatterjee, R. N. Banavar, "Robust Discrete-Time Pontryagin Maximum Principle on Matrix Lie Groups", IEEE Transactions on Automatic Control, 2021
- A. A. Joshi**, K. Subbarao, "Uncertainty Quantification and Analysis of Dynamical Systems with Invariants", AAS-19-368, 29th AAS/AIAA Space-Flight Mechanics Meeting, January, 2019

Technical Skills

PROGRAMMING: Python, PyTorch, MATLAB (Simulink, Control System Toolbox, Optimization Toolbox), C++

CONTROL & ESTIMATION: PID, LQR, Model Predictive Control (MPC), Sensor Fusion, State Estimation, Kalman Filters (EnKF, EKF, UKF)

COURSEWORK: Optimization, Reinforcement Learning, Machine Learning for Signal Processing, Deep Generative Models, State Estimation, Adaptive Control, Optimal Control, Stochastic Control, Probability and Random Processes, Stochastic Differential Equations